

Tug-of-War Tuning: Capacity and Cross-Lingual Transfer in Zero-Shot NER

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Abstract

We investigate how the amount of fine-tuned capacity affects zero-shot cross-lingual transfer for named entity recognition (NER). Training exclusively on English WikiANN and evaluating on 14 typologically diverse target languages, we find a consistent *inverted-U* pattern: cross-lingual transfer is worst at both ends of the parameter-efficiency spectrum. Full fine-tuning, which updates all parameters, transfers worst overall and collapses on non-Latin-script languages, while the most parameter-efficient methods, (IA)³ and prefix tuning, also underperform. Methods with an intermediate number of trainable parameters (Low-Rank Adaptation (LoRA; Hu et al., 2022) and Housby adapters) transfer best, with LoRA outperforming full fine-tuning by roughly 7 F1 points on average and by over 10 points on languages such as Chinese and Japanese. We find the same inverted-U shape within LoRA alone: a sweep over ranks shows cross-lingual transfer peaking at an intermediate rank, distinct from the higher rank that maximises in-language English performance, revealing a rank-transfer tradeoff. We also examine, and do not find clean support for, the common assumption that better preservation of cross-lingual representational alignment explains PEFT’s transfer benefits: the method that best preserves a standard alignment measure transfers worst, while LoRA’s larger alignment shift coincides with its best transfer. Given known limitations of this alignment measure, we present this as a preliminary, metric-dependent observation that motivates further analysis with more robust alignment metrics, rather than a settled claim.

1 Introduction

Cross-lingual transfer from high-resource to low-resource languages is a central challenge in multilingual NLP (Pires et al., 2019; Wu and Dredze, 2019). A standard approach fine-tunes a multilingual pre-trained model on a source language and

evaluates zero-shot on target languages (Conneau et al., 2020). Full fine-tuning updates every weight in the model and is known to cause *catastrophic forgetting* (McCloskey and Cohen, 1989) of multilingual structure, degrading transfer especially to typologically and orthographically distant languages.

Parameter-efficient fine-tuning (PEFT) methods, including LoRA (Hu et al., 2022), Housby adapters (Housby et al., 2019), (IA)³ (Liu et al., 2022), and prefix tuning (Li and Liang, 2021), update only a small fraction of parameters and are often motivated as a way to preserve multilingual structure during fine-tuning (Ansell et al., 2022; Üstün et al., 2022). This suggests a natural hypothesis: the fewer parameters a method updates, the better it should preserve cross-lingual transfer. Our results complicate this picture in two ways.

First, transfer does not improve monotonically as fine-tuned capacity shrinks. Across five methods spanning nearly the full parameter-efficiency spectrum, we observe an inverted-U: full fine-tuning (100% of parameters) transfers worst, the most parameter-efficient methods, (IA)³ and prefix tuning, also underperform, and two methods with an intermediate number of trainable parameters, LoRA and Housby adapters, transfer best. We find the identical pattern within LoRA alone: sweeping the rank r , which controls how many parameters LoRA updates (a lower r means fewer trainable parameters and a more constrained, more parameter-efficient update, while a higher r allows a richer but less constrained one), shows cross-lingual transfer rising with r up to a point and then declining, at a lower rank than the one that maximises English-only performance.

Second, we ask whether a standard proxy for “preserving multilingual structure” (cosine similarity between source- and target-language representations) actually predicts this pattern. It does not: the method with the smallest average change in this

measure, prefix tuning, has the worst transfer, while LoRA, which shows the largest change, has the best. Taken at face value, this would suggest that methods which perturb representations more are, if anything, more transfer-preserving: the opposite of what the alignment-preservation motivation for PEFT would predict. However, mean-pooled cosine similarity is known to be a coarse, anisotropy-confounded proxy for cross-lingual structure (Cai et al., 2021; Vulić et al., 2020), so we treat this second finding as suggestive rather than conclusive, and offer a subspace-based hypothesis (§4.3, Appendix A) for why a method could shift this metric substantially while still preserving the narrower set of directions that matter for the NER task.

We study both patterns on WikiANN (Pan et al., 2017) (15 languages, 9 language families) using XLM-RoBERTa base (Conneau et al., 2020). Our contributions are:

- A controlled, three-seed comparison of five fine-tuning methods spanning the parameter-efficiency spectrum, showing a consistent inverted-U relationship with zero-shot cross-lingual NER transfer;
- A rank sweep over LoRA ($r \in \{1, \dots, 768\}$) showing the same inverted-U shape within a single method, with the optimal rank for cross-lingual transfer lower than the optimal rank for the English source language;
- A preliminary examination of whether representational alignment (via mean-pooled cosine similarity) explains these patterns, finding that it does not straightforwardly, along with a subspace-decomposition account for why a coarse alignment metric may be a poor predictor of task-relevant transfer.

2 Background and Related Work

PEFT for cross-lingual transfer. Prior work motivates PEFT as beneficial for cross-lingual transfer by limiting how much a fine-tuning step can disturb pretrained multilingual representations: Üstün et al. (2022) found Housby adapters outperform full fine-tuning for low-resource dependency parsing transfer, and Pfeiffer et al. (2020) proposed MAD-X, which composes language- and task-specific adapters to keep cross-lingual representations largely intact. Ansell et al. (2022) pursue a similar goal via composable sparse fine-tuning. These works establish that some PEFT methods

outperform full fine-tuning cross-lingually, but do not compare across a range of parameter budgets or examine whether increasingly parameter-efficient methods keep improving transfer. Lialin et al. (2023) and Chen et al. (2022) study how LoRA’s rank affects performance, but in monolingual settings; we are not aware of prior work characterising a rank-transfer tradeoff specifically for cross-lingual transfer. Yadav (2026) report related evidence of representational collapse in code-mixed generation, consistent with the idea that both too much and too little constraint on fine-tuning can distort multilingual representations. BitFit (Zaken et al., 2022), Compacter (Mahabadi et al., 2021), and AdaLoRA (Zhang et al., 2023) occupy additional points on the PEFT parameter-efficiency spectrum; AdaLoRA’s rank-pruning objective, which adaptively reallocates rank across layers, is a natural comparison against our fixed-rank sweep (§5).

Representational alignment and its limitations.

A common way to quantify how much fine-tuning disturbs cross-lingual structure is mean-pooled cosine similarity between source- and target-language sentence representations (Wu et al., 2020; Zhao et al., 2021). This measure has known weaknesses, however: retrieval-based metrics such as BLI precision@1 and CSLS (Conneau et al., 2018) correlate more reliably with downstream cross-lingual performance (Vulić et al., 2020; Ravishankar et al., 2019), and CKA (Kornblith et al., 2019) is more robust to isotropic scaling. Most importantly for our analysis, Cai et al. (2021) show that mean-pooled cosine similarity is confounded by representation anisotropy, so a high similarity score can reflect a general “cone collapse” of the representation space rather than genuine cross-lingual alignment. We use mean-pooled cosine similarity in §4.3 for direct comparability with this prior work, but given these confounds, we treat any conclusions drawn from it as preliminary (see Limitations).

3 Experimental Setup

Model and data. We fine-tune the pretrained XLM-R base checkpoint (Conneau et al., 2020) (125M parameters; this checkpoint has not been previously fine-tuned on any NER data) on the English portion of WikiANN (Pan et al., 2017) (entity types PER, ORG, LOC) and evaluate zero-shot, with no target-language supervision, on 14 additional languages: ES, FR, DE, ZH, AR, HI, RU,

JA, FI, SW, TR, KO, NL, VI, spanning 9 language families and 5 writing scripts.

Fine-tuning methods. We compare five methods that update different amounts of the underlying model, presented from most to least trainable parameters. **Full fine-tuning** updates every weight matrix in the model directly. **LoRA** (Hu et al., 2022) instead freezes a pretrained weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d \times k}$ (with d and k the input and output dimensions of a given projection) and learns a low-rank update:

$$\begin{aligned} \mathbf{W} &= \mathbf{W}_0 + \mathbf{B}\mathbf{A}, \\ \mathbf{B} &\in \mathbb{R}^{d \times r}, \mathbf{A} \in \mathbb{R}^{r \times k}, r \ll \min(d, k) \end{aligned} \quad (1)$$

The rank r is a hyperparameter that controls how many parameters LoRA updates: a lower r means fewer trainable parameters (a more constrained, more parameter-efficient update), while a higher r allows a richer, less constrained update that approaches full fine-tuning as $r \rightarrow \min(d, k)$. We apply LoRA to the query and value projections of all 12 attention layers ($r=16$, $\alpha=32$, dropout 0.1; 0.60M trainable parameters, 0.47% of the model). Key and feed-forward projections are frozen; layer-wise ablations are underway (§5).

Houlsby adapters (Houlsby et al., 2019) insert bottleneck modules after *both* the self-attention and feed-forward sub-layers in all 12 layers, transforming a hidden state $\mathbf{h} \in \mathbb{R}^d$ as:

$$\mathbf{h}' = \mathbf{h} + \mathbf{W}_{\text{up}} \sigma(\mathbf{W}_{\text{dn}} \mathbf{h}), \quad (2)$$

with $\mathbf{W}_{\text{dn}} \in \mathbb{R}^{b \times d}$, $\mathbf{W}_{\text{up}} \in \mathbb{R}^{d \times b}$, bottleneck dimension $b=64$, and σ the GeLU nonlinearity. (IA)³ (Liu et al., 2022) rescales hidden states element-wise ($\mathbf{h}' = \mathbf{1} \odot \mathbf{h}$, $\mathbf{1} \in \mathbb{R}^d$) on keys, values, and feed-forward intermediates in all 12 layers (0.030M trainable parameters, 0.024%). **Prefix tuning** (Li and Liang, 2021) prepends $m=10$ trainable tokens to each layer’s key/value attention while keeping all original weights frozen.

Training details. All methods are trained with AdamW (Loshchilov and Hutter, 2019), a 10% warmup schedule, weight decay 0.01, for 3 epochs with batch size 32. We select the learning rate per method based on English validation F1: 2×10^{-4} for full fine-tuning, LoRA, and Houlsby adapters; 3×10^{-3} for (IA)³; 1×10^{-2} for prefix tuning. All results are reported as mean $\pm$ std over three random seeds ({42, 123, 456}); we use $n=3$ throughout to refer to this seed count. Significance is assessed

with paired t -tests ($df=2$) per language; given the exploratory scope and small n , we do not apply a multiple-comparison correction, and we treat significance at this sample size as suggestive rather than confirmatory.

Alignment measurement. Following Wu et al. (2020), for each target language $\mathcal{L}$ we compute alignment as the mean cosine similarity between ℓ_2 -normalised, mean-pooled representations of $N=1,000$ FLORES-200 (NLLB Team et al., 2022) devtest sentence pairs (the same sample is used for every method). For each sentence, we mean-pool the final transformer layer’s hidden states over non-padding tokens, then ℓ_2 -normalise; we use only this final layer throughout, so no per-layer index is needed. Writing $\mathbf{h}_i^{\text{EN}}$ and $\mathbf{h}_i^{\mathcal{L}}$ for the resulting representations of the i -th sentence pair ($i = 1, \dots, N$) in English and language $\mathcal{L}$:

$$\text{Align}(\mathcal{L}) = \frac{1}{N} \sum_{i=1}^N \frac{\mathbf{h}_i^{\text{EN}} \cdot \mathbf{h}_i^{\mathcal{L}}}{\|\mathbf{h}_i^{\text{EN}}\| \|\mathbf{h}_i^{\mathcal{L}}\|} \quad (3)$$

We define alignment degradation as $\Delta_{\mathcal{L}} = \text{Align}_{\text{base}}(\mathcal{L}) - \text{Align}_{\text{ft}}(\mathcal{L})$, where $\text{Align}_{\text{base}}(\mathcal{L})$ is computed on the pretrained model before any fine-tuning and $\text{Align}_{\text{ft}}(\mathcal{L})$ after fine-tuning with a given method. Under this definition, $\Delta_{\mathcal{L}} > 0$ indicates that fine-tuning *reduced* alignment relative to the pretrained model, so a larger $\Delta_{\mathcal{L}}$ means worse preservation, and a value near 0 means alignment was essentially unchanged. No anisotropy correction is applied; absolute alignment values near 0.99 are expected under this uncorrected measure (Wu et al., 2020), and we only interpret the relative $\Delta_{\mathcal{L}}$ across methods, not its absolute magnitude. Adapter alignment measurement is ongoing and excluded from Table 3.

4 Results

4.1 Main Transfer Results: A Parameter-Efficiency Sweet Spot

Table 1 reports span-level F1. Ordering the five methods by how many parameters they update (full fine-tuning (100%) > Houlsby adapters > LoRA (0.47%) > (IA)³ (0.024%) > prefix tuning), mean cross-lingual F1 traces an inverted U rather than a monotonic trend: full fine-tuning transfers worst (51.8%), performance rises through Houlsby adapters (58.6%) and peaks at LoRA (58.9%, a ~ 7.1 pp gap over full fine-tuning), then falls again

L	Fam.	FT	Adp	LoRA	IA ³	Pfx
en	Gmc	80.5	77.3	76.2	75.9	74.7
es	Rom	62.2	69.2	69.9	65.4	61.9
fr	Rom	71.1	72.7	73.0	70.4	65.6
de†	Gmc	67.3	71.3	71.2	70.1	66.0
nl†	Gmc	75.1	76.7	76.8	76.4	72.6
ru	Slv	55.4	58.6	58.0	57.0	46.3
fi†	Ura	70.1	72.0	71.4	71.9	66.9
sw	N-C	62.0	66.5	66.8	63.6	62.3
tr†	Trk	61.6	70.6	72.6	70.4	61.3
vi	A-A	60.9	63.0	65.5	60.2	60.6
zh†	S-T	13.1	26.1	26.9	21.5	19.6
ar	Sem	31.5	42.1	42.6	48.6	39.4
hi†	Ind	52.5	64.8	61.8	62.2	57.9
ja†	Jpn	10.2	20.2	21.4	16.4	14.6
ko†	Kor	32.7	46.3	46.1	46.4	41.0
Avg	—	51.8	58.6	58.9	57.2	52.6

Table 1: Span-level F1 (%) on WikiANN zero-shot NER, mean over 3 seeds (full std in Appendix C). † = LoRA significantly outperforms full FT ($p < 0.05$). The 0.3 pp LoRA–adapter gap is not statistically significant. Bold = best per row. Fam.: Gmc Germanic, Rom Romance, Slv Slavic, Ura Uralic, N-C Niger-Congo, Trk Turkic, A-A Austroasiatic, S-T Sino-Tibetan, Sem Semitic, Ind Indic, Jpn Japonic, Kor Koreanic.

through (IA)³ (57.2%) to prefix tuning, the second-worst (52.6%). LoRA significantly outperforms full fine-tuning ($p < 0.05$, paired t -test) in 8 of the 14 target languages, concentrated in typologically distant and non-Latin-script languages. The gap between LoRA and Housby adapters (0.3 pp) is small relative to seed variance and not significant at $n=3$.

Full fine-tuning collapses most severely on non-Latin-script languages: Chinese reaches only $13.1 \pm 3.6\%$ F1 under full fine-tuning versus $26.9 \pm 0.5\%$ under LoRA, and Japanese only $10.2 \pm 2.4\%$ versus $21.4 \pm 1.3\%$. Full fine-tuning’s high variance relative to LoRA’s stability on these languages suggests a high-variance, low-performance regime produced by unconstrained updates interacting with scripts that are less represented during pretraining.

This advantage is not limited to non-Latin scripts, however: Turkish, which uses the Latin script but is morphologically rich and typologically distant from English (agglutinative morphology, flexible word order), shows one of the largest gaps between LoRA and full fine-tuning (72.6% vs. 61.6%, +11.0 pp). This suggests the collapse under full parameter updates tracks typological distance more broadly, not simply script mismatch,

a distinction we did not draw clearly in an earlier draft, which grouped Turkish together with Chinese and Japanese under a single “script-distant” label.

At the other end of the spectrum, (IA)³, despite updating far fewer parameters than LoRA, outperforms LoRA on Arabic and Hindi, suggesting that for some language families, rescaling-only updates preserve relevant features about as well as, or better than, a low-rank update. Housby adapters are the most stable method overall (average std ± 0.74) and outperform LoRA on Hindi (64.8 vs. 61.8).

4.2 A Rank–Transfer Tradeoff Within LoRA

If the inverted-U pattern above reflects fine-tuned capacity broadly, rather than something specific to comparing different methods, the same shape should appear when varying capacity within a single method. Figure 1 shows English and cross-lingual F1 as a function of LoRA rank $r \in \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 768\}$, together with the corresponding trainable parameter count on a secondary axis. Both F1 curves rise together up to $r=64$; beyond that point they diverge. English F1 continues rising to its maximum at $r=768$ (81.3%), the highest-capacity setting we test, while cross-lingual F1 peaks earlier, at $r=256$ (61.2%), and then declines:

$$\begin{aligned} r_{\text{EN}}^* &= \arg \max_r \mathcal{F}_{\text{EN}}(r) = 768 \\ r_{\text{xling}}^* &= \arg \max_r \mathcal{F}_{\text{xling}}(r) = 256 \end{aligned} \quad (4)$$

This mirrors §4.1: just as the highest-capacity method (full fine-tuning) transfers worse than an intermediate-capacity one (LoRA), the highest-rank LoRA setting transfers worse than an intermediate rank. The decline beyond $r=256$ is concentrated in the same typologically distant languages implicated above: at $r=768$, Arabic falls from 50.6 (at $r=8$) to 39.4, and Chinese falls from 31.1 (at $r=256$) to 23.1, the same failure pattern seen in full fine-tuning. Full numeric results across all eleven ranks are provided in Appendix B (Table 4); Table 5 confirms the decline is consistent across seeds for four representative ranks, not a single-run artefact, with per-rank standard deviations (≤ 0.8 pp) well below the 2.5 pp gap between $r=256$ and $r=768$ (Fig. 1).

Practically, this means the rank that is best for the source language is not the rank that is best for cross-lingual transfer: a practitioner tuning LoRA rank

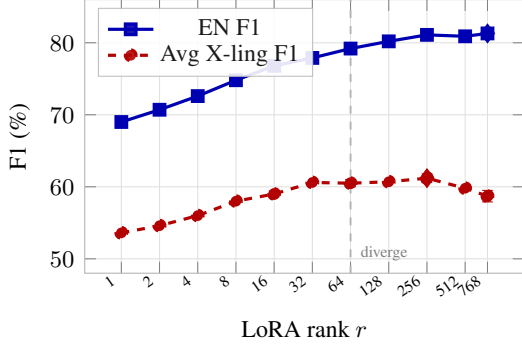


Figure 1: EN F1 and average cross-lingual F1 vs. LoRA rank r ($\log_2$ scale). Diamonds mark the optima ($r_{\text{xling}}^*=256$, $r_{\text{EN}}^*=768$). Error bars (± 1 std over 3 seeds; Table 5) are shown at the four seed-verified ranks; the $r=256$ -to- $r=768$ cross-lingual decline exceeds this seed variability. The trainable parameter count (Table 4) grows monotonically with rank, making clear that the cross-lingual optimum is not the highest-capacity setting available.

purely on English validation performance would select $r=768$ and lose 2.5 points of cross-lingual F1 relative to $r=256$.

4.3 Does Representational Alignment Explain This Pattern?

A natural explanation for the inverted-U in §4.1 would be that intermediate-capacity methods better preserve cross-lingual representational alignment, and that this preserved alignment is what drives better transfer. Table 2 tests this directly using the mean-pooled cosine alignment measure from §3 (we report full fine-tuning, LoRA, (IA)³, and prefix tuning; Houlsby adapter alignment is not yet computed, see Limitations).

Method	$\bar{\Delta} \downarrow$	Xling F1 $\uparrow$
Prefix	0.005	52.6
(IA) ³	0.038	57.2
Full FT	0.055	51.8
LoRA	0.061	58.9

$\downarrow$ =better preserved $\uparrow$ =better transfer

Table 2: Alignment degradation vs. cross-lingual transfer, averaged across languages.

This explanation does not hold cleanly. If anything, alignment preservation and transfer are inversely related across these four methods: prefix tuning preserves alignment almost perfectly ($\bar{\Delta}=0.005$) yet has the second-worst transfer, while LoRA shows the largest average shift ($\bar{\Delta}=0.061$) yet transfers best. Table 3 (per-language) shows this holds language by language, including for Chinese, where LoRA shifts alignment more than full

fine-tuning does (0.075 vs. 0.048) while still outperforming it by 13.7 pp in F1.

Lang	FT	LoRA	IA ³	Pfx
es	.038	.045	.046	.003
fr	.030	.050	.037	.003
de	.048	.050	.032	.004
zh	.048	.075	.042	.005
ar	.074	.064	.047	.007
hi	.075	.063	.037	.007
ru	.035	.063	.031	.005
ja	.065	.069	.035	.005
fi	.065	.067	.037	.006
sw	.074	.072	.046	.006
tr	.082	.064	.037	.006
ko	.082	.069	.037	.007
nl	.026	.046	.036	.004
vi	.035	.054	.028	.005
Avg $\bar{\Delta}$	.055	.061	.038	.005

Table 3: Alignment degradation $\Delta_{\mathcal{L}}$ (Eq. 3). Adapter scores not yet computed. Lower = better preserved. No anisotropy correction; see §2–3.

We do not read this as evidence that disturbing representational alignment *helps* transfer. Mean-pooled cosine similarity is a coarse, whole-representation measure known to be confounded by representation anisotropy: a change in this score can reflect a broad rotation or rescaling of the representation space that has little to do with the narrower set of directions a downstream task actually relies on (Cai et al., 2021). Appendix A sketches one way this could happen: if a method’s update is concentrated outside the task-relevant subspace, it can shift mean-pooled cosine substantially while leaving task-relevant directions largely intact: LoRA’s low-rank update is a natural candidate, since by construction it can only act within an r -dimensional subspace of the representation space. Conversely, a method that freezes all weights (prefix tuning) trivially preserves the whole-representation measure but also cannot re-route task-relevant directions if they need adjusting.

We want to be careful about how much weight this analysis can bear. Because our alignment measure has known limitations, we cannot rule out that a more targeted metric (CKA restricted to task-relevant directions, or retrieval-based measures like CSLS) would show a relationship more consistent with the standard alignment-preservation account. We therefore present this analysis as a preliminary, metric-dependent observation and a motivation for the subspace hypothesis in Appendix A, rather than as a robust finding that alignment preservation is unrelated to, or actively harmful for, transfer.

5 Discussion

Practical guidance. For cross-lingual NER with XLM-R base, our results suggest treating parameter efficiency as a dial with a sweet spot, rather than a scale where “more efficient is always safer”: both full fine-tuning and the most parameter-efficient methods we test underperform intermediate-capacity methods. Within LoRA specifically, rank $r \in [32, 256]$ gives the best transfer, with $r \leq 64$ preferable when the deployment targets typologically distant languages such as Chinese, Japanese, Korean, or Turkish, where the gap between LoRA and full fine-tuning is largest and most stable. Houlsby adapters are a reasonable alternative when rank tuning is impractical, given their stability across languages (58.6% average, std ± 0.74).

Limitations. Our central inverted-U finding rests on five methods and one encoder (XLM-R base) with English as the sole source language; whether it holds for larger encoders, decoder-only models, or other source languages is future work. Because LoRA, Houlsby adapters, (IA)³, prefix tuning, and full fine-tuning differ in *where* and *how* they modify the model (Eq. 1, Eq. 2), not only in how many parameters they update, our comparison confounds trainable-parameter count with adaptation architecture, and parameter count alone cannot be identified as the sole driver of the inverted-U pattern. WikiANN is silver-annotated, and we plan to validate our main comparison on MultiCoNER and UNER v2. Adapter alignment scores are not yet computed. Most importantly, our alignment analysis (§4.3) relies on mean-pooled cosine similarity, a measure we ourselves note is confounded by representation anisotropy and less predictive of downstream transfer than retrieval-based or CKA-based alternatives; we chose it for direct comparability with prior work using the same measure (Wu et al., 2020), but this means our alignment-related conclusions should be read as tentative and specific to this metric, not as a general claim about the relationship between representational alignment and cross-lingual transfer. We do not evaluate BitFit, Compacter, or AdaLoRA; AdaLoRA’s rank-pruning mechanism is a natural point of comparison against our fixed-rank sweep.

Future work. Layer-wise ablations (zeroing LoRA’s update per layer and measuring the resulting drop in cross-lingual F1) are underway; pre-

liminary results suggest layers 0–5 carry disproportionate transfer signal for typologically distant languages, which would directly test the subspace hypothesis in Appendix A. We plan to compute CKA restricted to task-relevant directions (Kornblith et al., 2019) and retrieval-based diagnostics (CSLS, BLI P@1; Vulić et al., 2020; Conneau et al., 2018) to assess whether a more targeted alignment measure better predicts the inverted-U pattern than mean-pooled cosine similarity, and we will extend all alignment measurements to adapters.

6 Conclusion

We present a controlled, five-way comparison of fine-tuning methods spanning the parameter-efficiency spectrum for zero-shot cross-lingual NER across 14 languages. Rather than transfer improving monotonically as fine-tuned capacity shrinks, we find an inverted-U: full fine-tuning and the most parameter-efficient methods, (IA)³ and prefix tuning, both underperform intermediate-capacity methods (LoRA, Houlsby adapters), with LoRA significantly outperforming full fine-tuning in 8 of 14 languages, especially typologically and orthographically distant ones. A rank sweep shows the same shape within LoRA alone, revealing a rank–transfer tradeoff in which the rank that maximises English performance is not the rank that maximises cross-lingual transfer. We additionally examine, using a standard but limited alignment metric, whether representational alignment preservation explains this pattern, and find that it does not straightforwardly, a preliminary result we discuss cautiously given the metric’s known confounds, alongside a subspace-based hypothesis for reconciling it with the main inverted-U finding.

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A A Subspace Account of the Alignment–Transfer Relationship

We sketch why a method’s effect on global cosine alignment need not track its effect on cross-lingual task performance, and how this could be consistent with the inverted-U pattern in §4.1–§4.2.

Setup. Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ be the encoder. Partition the representation space as $\mathbb{R}^d = \mathcal{S}_\tau \oplus \mathcal{S}_\tau^\perp$, where $\mathcal{S}_\tau$ is the *task-relevant subspace* (directions encoding entity-boundary features shared across languages) and $\mathcal{S}_\tau^\perp$ its complement. Any hidden state decomposes as $\mathbf{h} = \mathbf{h}_\tau + \mathbf{h}_\perp$.

Update decomposition. A weight update $\Delta\mathbf{W}$ acts as $\mathbf{h} \mapsto \mathbf{h} + \Delta\mathbf{W}\mathbf{h}$. Decomposing:

$$\Delta\mathbf{W}\mathbf{h} = \underbrace{\Delta\mathbf{W}\mathbf{h}_\tau}_{\text{task subsp.}} + \underbrace{\Delta\mathbf{W}\mathbf{h}_\perp}_{\text{orth. compl.}} \quad (5)$$

Too little capacity: prefix tuning. All original weights are frozen ($\Delta\mathbf{W}=\mathbf{0}$), so $\Delta_{\mathcal{L}}^{\text{pfx}} \approx 0$ (observed: $\hat{\Delta}=0.005$):

$$\Delta_{\mathcal{L}}^{\text{pfx}} = \text{Align}_{\text{base}}(\mathcal{L}) - \text{Align}_{\text{pfx}}(\mathcal{L}) \approx 0. \quad (6)$$

Soft prompts modulate attention but cannot directly re-route directions within $\mathcal{S}_\tau$ if the task requires it, consistent with prefix tuning’s weak transfer despite near-perfect alignment preservation.

Too much capacity: full fine-tuning. Under an isotropy assumption (Zhu et al., 2021), the expected interference of an unconstrained update with the task subspace scales as:

$$\mathbb{E}[\|\Pi_{\mathcal{S}_\tau} \Delta\mathbf{W}\|_F^2] \propto \frac{\dim(\mathcal{S}_\tau)}{d} \quad (7)$$

An unconstrained, full-rank update has no built-in mechanism preventing it from acting on $\mathcal{S}_\tau$, giving it more opportunity to disturb task-relevant directions than a rank-constrained update.

An intermediate regime: LoRA. From Eq. (1), $\Delta\mathbf{W} = \mathbf{B}\mathbf{A}$ has rank r , so $\Delta\mathbf{W}\mathbf{h}$ lies in a subspace of dimension at most $r \ll d$:

$$\Delta\mathbf{W}\mathbf{h} \in \text{col}(\mathbf{B}), \quad \dim(\text{col}(\mathbf{B})) \leq r \ll d. \quad (8)$$

When $\text{col}(\mathbf{B})$ is not aligned with $\mathcal{S}_\tau$, plausible when r is small relative to $\dim(\mathcal{S}_\tau)$, the task component $\mathbf{h}_\tau$ can be largely preserved even though the update induces a large rotation of the complement, which is exactly what a whole-representation measure like mean-pooled cosine similarity picks up:

$$\Delta_{\mathcal{L}}^{\text{LoRA}} \approx \frac{\|\text{proj}_{\text{col}(\mathbf{B})} \bar{\mathbf{h}}^{\text{EN}}\|}{\|\bar{\mathbf{h}}^{\text{EN}}\|} \cdot \alpha_{\mathbf{A}}, \quad (9)$$

where $\bar{\mathbf{h}}^{\text{EN}}$ is the mean English representation and $\alpha_{\mathbf{A}}$ scales with $\mathbf{A}$. If $\text{col}(\mathbf{B})$ aligns with high-variance directions outside $\mathcal{S}_\tau$, $\Delta_{\mathcal{L}}^{\text{LoRA}}$ can be large even when $\mathcal{S}_\tau$ is preserved, and representation anisotropy (Cai et al., 2021) amplifies this, since mean-pooled cosine is dominated by a small number of high-variance directions regardless of their task relevance. This offers one account of why LoRA can show a larger $\hat{\Delta}$ than full fine-tuning while still transferring better: the rank constraint concentrates the update’s effect into a subspace that need not overlap with $\mathcal{S}_\tau$, rather than acting on all of $\mathbb{R}^d$ indiscriminately.

This account also suggests why a sweet spot exists at all: at very low rank (or under prefix tuning/(IA)³’s limited parameterisations), $\text{col}(\mathbf{B})$ or its analogue may be too constrained to include the directions in $\mathcal{S}_\tau$ that genuinely need adjusting, hurting transfer through insufficient capacity; at very high rank (approaching full fine-tuning), the update is no longer confined to a small subspace and has more opportunity to interfere with $\mathcal{S}_\tau$ under the isotropy argument above, hurting transfer through excess capacity. An intermediate rank can thread this needle: expressive enough to adjust the necessary directions in $\mathcal{S}_\tau$, constrained enough to avoid broadly interfering with it.

Proposition 1 (Alignment–Transfer Decoupling). *There exist configurations $(\mathbf{B}, \mathbf{A})$ such that the LoRA update simultaneously (i) increases global cosine degradation beyond full fine-tuning’s ($\Delta_{\mathcal{L}}^{\text{LoRA}} > \Delta_{\mathcal{L}}^{\text{FT}}$) and (ii) better preserves $\mathcal{S}_\tau$, yielding higher cross-lingual F1.*

Remark 1. Proposition 1 formalises why global cosine similarity need not track cross-lingual task

performance, and offers one possible mechanism for the inverted-U pattern in §4.1–§4.2, though we stress (§4.3) that we have not directly measured S_τ or verified this mechanism empirically. CKA (Kornblith et al., 2019) restricted to S_τ would help disentangle global from task-subspace geometry, and retrieval-based metrics (CSLS, BLI P@1) are less susceptible to anisotropy confounds (Vulić et al., 2020), motivating both as planned analyses.

B Rank Sweep: Full Results and Seed Variance

Table 4 reports the full rank sweep (single run per rank) underlying Figure 1. Table 5 confirms the key qualitative pattern (the optimal rank diverging between English and cross-lingual performance) is stable across three seeds for four representative ranks.

Rank	Params (M)	EN F1	Xling F1
1	0.042	69.0	53.6
2	0.079	70.7	54.6
4	0.153	72.6	56.0
8	0.300	74.8	58.0
16	0.595	76.8	59.0
32	1.185	77.9	60.6
64	2.365	79.2	60.5
128	4.724	80.2	60.7
256	9.443	81.1	61.2
512	18.880	80.9	59.8
768	28.317	81.3	58.7

Table 4: LoRA rank sweep (single run; seed variance in Table 5 below). Bold = column best. $r_{\text{xling}}^* = 256 \neq r_{\text{EN}}^* = 768$ (Eq. 4).

Rank	EN F1		Avg X-ling F1	
	Mean	Std	Mean	Std
16	76.8	0.6	59.0	0.4
64	79.2	0.5	60.5	0.5
256	81.1	0.4	61.2	0.6
768	81.3	0.3	58.7	0.8

Table 5: Rank sweep F1, mean±std over 3 seeds. The cross-lingual decline at $r=768$ vs. $r=256$ (−2.5 pp) is consistent across seeds, confirming Eq. 4.

C Full Per-Language Results with Standard Deviations

Lang	Full FT	Adp	LoRA	IA ³	Pfx
en	80.5±.1	77.3±.2	76.2±.3	75.9±.1	74.7±.3
es	62.2±2.4	69.2±.4	69.9 ±2.8	65.4±.7	61.9±.4
fr	71.1±.9	72.7±.2	73.0 ±.8	70.4±.1	65.6±.3
de	67.3±.6	71.3 ±.3	71.2±.2	70.1±.2	66.0±.4
nl	75.1±.2	76.7±.2	76.8 ±.4	76.4±.3	72.6±.2
ru	55.4±1.0	58.6 ±.95	58.0±.8	57.0±.6	46.3±.5
fi	70.1±.4	72.0 ±.4	71.4±.2	71.9±.4	66.9±.3
sw	62.0±1.3	66.5±.3	66.8 ±1.1	63.6±.3	62.3±.4
tr	61.6±1.3	70.6±.8	72.6 ±2.7	70.4±.8	61.3±.7
vi	60.9±1.6	63.0±2.4	65.5 ±1.7	60.2±.0	60.6±.3
zh	13.1±3.6	26.1±.9	26.9 ±.5	21.5±.1	19.6±.8
ar	31.5±5.7	42.1±1.9	42.6±3.3	48.6 ±1.9	39.4±.6
hi	52.5±2.1	64.8 ±.5	61.8±1.1	62.2±.8	57.9±.4
ja	10.2±2.4	20.2±.2	21.4 ±1.3	16.4±1.0	14.6±.6
ko	32.7±1.4	46.3±.95	46.1±.7	46.4 ±.2	41.0±.5
Avg	51.8	58.6	58.9	57.2	52.6

Table 6: Full per-language F1 (%) with mean±std over 3 seeds. Bold = best cross-lingual result per row. Multi-seed adapter average: 58.59 ± 0.74 . The 0.3 pp LoRA–adapter gap is not statistically significant.